

VOROZHEYKA, Russian Federation

WMO: 292530

Lat: 58.456N

Lon: 89.469E

Elev: 439

StdP: 14.46

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.7	-35.6	-45.6	0.4	-40.7	-39.6	0.6	-34.4	16.5	12.9	13.8	13.4	0.1	90	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	82.9	67.3	79.7	65.2	76.7	64.0	70.6	79.0	68.2	76.2	66.0	73.8	2.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.6	102.9	74.9	65.2	94.6	72.6	62.9	87.2	70.2	34.8	79.3	32.8	76.1	31.1	74.0	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	10.0	8.5	DB	-48.7	88.0	7.0	3.3	-53.7	90.4	-57.8	92.4	-61.7	94.3	-66.8	96.7	(4)
(5)			WB	-47.3	73.3	5.5	2.9	-51.3	75.3	-54.5	77.0	-57.6	78.6	-61.6	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.6	-5.8	2.6	18.5	32.8	46.7	60.0	63.7	57.8	46.2	31.8	11.0	-0.1		
	DBStd	26.66	17.26	14.52	12.18	10.10	9.96	7.85	5.91	6.36	7.89	10.20	16.56	17.48		
	HDD50	8220	1729	1327	975	519	188	15	0	11	165	566	1170	1555		
	HDD65	12669	2194	1747	1440	967	571	184	96	236	564	1030	1620	2020		
	CDD50	1133	0	0	0	2	86	314	424	253	52	2	0	0		
	CDD65	106	0	0	0	0	4	33	55	13	1	0	0	0		
	CDH74	1259	0	0	0	1	76	475	539	153	15	0	0	0		
	CDH80	240	0	0	0	0	10	106	101	23	0	0	0	0		
(14) Wind	WSAvg	2.5	2.1	2.2	2.9	3.3	3.2	2.2	1.7	1.9	2.6	2.8	2.7	2.6		
(15) Precipitation	PrecAvg	19.7	1.1	0.9	0.9	1.2	1.7	2.1	2.0	2.3	2.2	2.0	1.9	1.7		
	PrecMax	27.2	2.9	1.7	1.9	2.6	4.1	5.2	4.4	5.3	5.3	3.8	3.2	3.3		
	PrecMin	13.5	0.0	0.4	0.1	0.3	0.5	0.0	0.3	0.7	0.6	0.6	0.6	0.2		
	PrecStd	3.3	0.6	0.4	0.4	0.6	1.0	1.1	0.9	1.3	1.1	0.7	0.6	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.4	33.3	47.6	64.0	80.9	87.6	86.9	83.2	76.1	60.8	41.8	33.3		
		MCWB	27.4	30.0	37.4	50.2	60.1	69.0	70.5	69.0	61.6	49.7	38.9	30.7		
	2%	DB	26.7	29.7	42.7	58.0	75.3	83.0	83.0	78.3	69.5	54.2	37.5	29.8		
		MCWB	25.2	27.3	34.6	45.9	56.9	66.3	69.0	64.6	58.1	45.8	34.7	28.2		
	5%	DB	23.1	26.5	38.9	53.2	70.3	79.6	79.6	74.5	64.4	49.3	34.6	26.1		
		MCWB	21.5	24.3	32.5	43.5	54.8	64.1	67.1	63.6	55.5	43.3	32.5	24.6		
	10%	DB	19.2	22.9	35.4	48.4	64.9	75.9	76.8	70.6	59.3	44.7	32.1	22.5		
		MCWB	18.0	21.0	31.0	40.2	51.8	62.4	65.7	61.8	52.6	41.1	30.4	21.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.9	30.6	39.9	51.9	61.8	72.2	74.4	71.2	64.0	50.5	39.7	31.4		
		MCDB	28.6	32.5	45.8	61.3	77.3	82.6	82.3	78.9	72.2	57.2	40.6	32.7		
	2%	WB	25.3	27.6	35.4	47.8	58.8	69.1	71.5	67.9	59.4	47.0	35.1	28.3		
		MCDB	26.7	29.5	40.2	56.2	71.8	79.6	79.2	74.7	67.0	53.1	36.6	29.6		
	5%	WB	21.7	24.4	33.3	43.8	56.1	66.4	69.3	65.3	56.7	43.9	32.6	24.7		
		MCDB	23.2	26.3	38.0	52.4	67.5	75.9	76.8	71.6	62.7	48.0	34.3	26.0		
	10%	WB	18.0	20.9	31.1	40.6	53.0	64.0	67.2	62.7	53.6	41.3	30.3	21.3		
		MCDB	19.4	23.1	35.6	47.3	63.8	72.9	74.8	68.8	59.0	44.5	32.3	22.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	18.6	20.4	18.5	19.7	21.0	20.1	18.9	16.6	12.0	13.0	13.8		
		MCDBR	12.9	14.9	22.6	25.2	28.5	28.5	25.0	25.8	26.4	19.9	9.9	13.5		
	5% WB	MCWBR	12.3	13.3	16.6	16.7	15.7	14.1	13.0	14.3	16.1	13.1	9.2	13.2		
		MCWBR	12.6	14.4	18.5	23.3	26.2	24.6	22.1	21.9	23.4	16.8	9.8	13.5		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.316	0.355	0.381	0.396	0.401	0.440	0.403	0.364	0.338	0.297	0.274		
	taud		1.968	1.970	1.946	2.055	2.247	2.321	2.198	2.292	2.360	2.246	2.014	1.919		
	Ebn at Noon		157	211	239	258	265	265	250	251	243	208	148	114		
	Edh at Noon		18	30	41	44	39	37	41	35	28	23	17	14		
(44) All-Sky Solar Radiation	RadAvg		189	482	918	1381	1547	1839	1706	1286	769	381	178	102		
	RadStd		24	39	74	124	162	194	114	121	115	40	36	20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page